MEEG432 Project 1: CFD - flow viz., flow field, and forces
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TASK 1: THE MODEL

This project will be analyzing the aerodynamics of the wing shown in Figure 1 below. This wing was chosen because the model was given to me by the course instructor. I also found it interesting to study a swept wing with a variable chord length.It is a swept wing that goes from a NACA 63515 root foil to a tip of NACA0016.

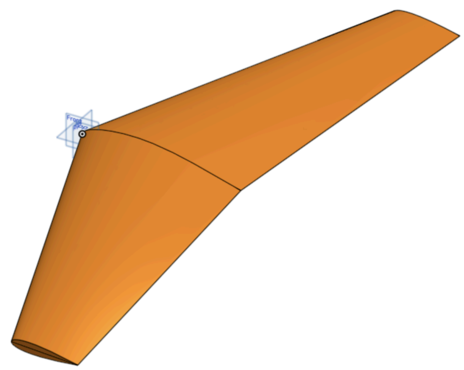

Figure 1: OnShape CAD model of the wing

The course instructor has provided the following wing characteristics: “The angle of attack is 4 degrees, the span is 10 m, the sweep angle is 20 degrees, and the root and tip chord are 1.5 m and 0.5 m respectively.” These seem to be incorrect values, as the correct dimensions are provided in the OnShape Drawing in Figure 2 below. Also included is the wing thickness at the root and the tip.

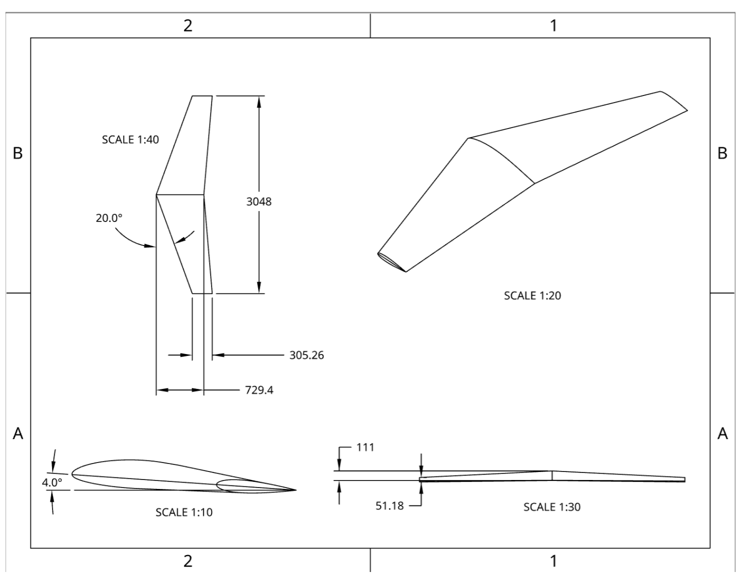

Figure 2: Dimensioned OnShape Drawing with an isometric, side, front, and top views

TASK 2: ASSESSING FLOW STATE

The Reynolds Number and the Mach number is needed to assess the flow state. The Reynolds Number will show if we can use a turbulent flow model to study the flow, while the Mach Number will show if we can assume the flow is incompressible. The Reynolds number is calculated with Equation 1 below, where ρ is the density of air, U_∞ is the flow velocity, L is the characteristic length, and μ is the dynamic viscosity of air. I will be using a flow velocity of 40 m/s. Also, because the chord length is variable across the span of the wing, I will be using the tip chord to be conservative. Values for the properties of air were found using the Engineering Toolbox at S.T.P.

$$Re = \frac{\rho U_\infty L}{\mu} \qquad [1]$$

$$Re = \frac{(1.292\ kg/m^3)(40\ m/s)(.30526\ m)}{17.15\times10^{-6}\ Pa\cdot s}$$

$$Re \approx 920,000 \gg 4,000 \text{ (turbulent)}$$

The Mach number is calculated with Equation 2 below, where c is the speed of sound in air.

$$Ma = \frac{U_\infty}{c} \qquad [2]$$

$$Ma = \frac{40\ m/s}{343}$$

$$Ma \approx 0.12 \ll 0.3 \text{ (incompressible)}$$

The calculations show that the flow is both turbulent and incompressible, which means we can use an incompressible flow solver with a turbulent flow model for these flow and geometric conditions.

TASK 3: FLOW VISUALIZATION

In order to figure out if the flow is attached and whether there are meaningful rotational structures, we can use the flow tracer tool on Simscale. Because the flow is steady, these tracers represent pathlines, streaklines, and streamlines, which are all equal in steady flow. I assume that if the flow was not steady, it would represent pathlines as it seems like the tracer is simulating a particle's path over the object. Figure 3 below shows the tracers down near the centerline and near the wingtip.

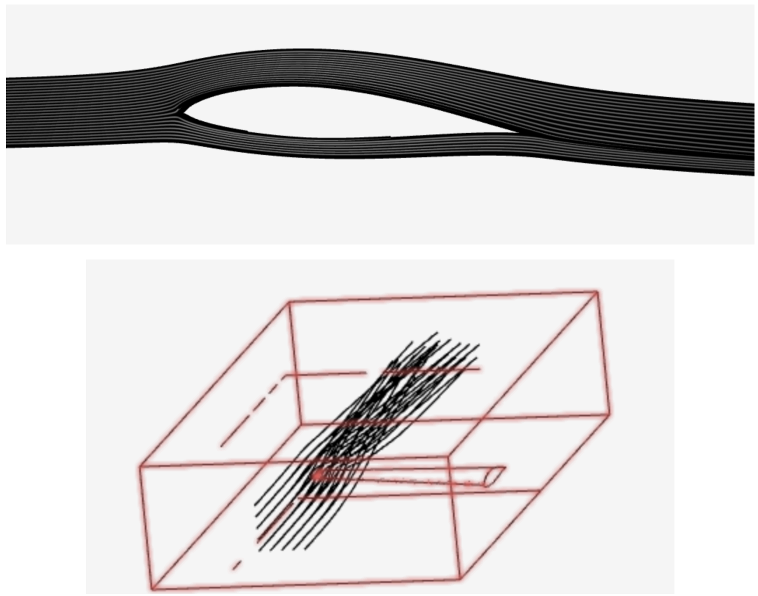

Figure 3: Side view of the flow tracer paths down the centerline (top)
Isometric view of flow tracer paths near the wingtip (bottom)

As shown above, the flow is attached near the centerline down the entire body. This is not true near the wingtip, however. The Kutta condition ensures there is continuity between the top and bottom surface where they connect at the wingtip. This causes the high pressure air below the wing to connect with the low pressure air above the wing which creates vortices and a stagnation point at the wing tip. This is because the flow needs circulation in order to satisfy the Kutta condition.

TASK 4: FLOW FIELD CONTOURS

Simscale also provides flow field contours to show flow properties such as velocity and pressure. Figure 4 below shows a side view of the flow velocity contours for X and Z down the centerline of the wing. For the Z-direction, there is a highly concentrated area of high positive velocity at the top surface by the leading edge that becomes an area of high negative velocity by the trailing edge. Also, there is a highly concentrated area of high negative velocity at the bottom surface by the leading edge that becomes slightly positive by the trailing edge. This can be seen just by observing the pathlines shown in the section above, where you can note when the path is sloped upwards versus being sloped downwards. In the X-direction, there are much faster velocities above the wing than below the wing. There is also apparent drag right on the surface of the wing, where the flow is much slower.

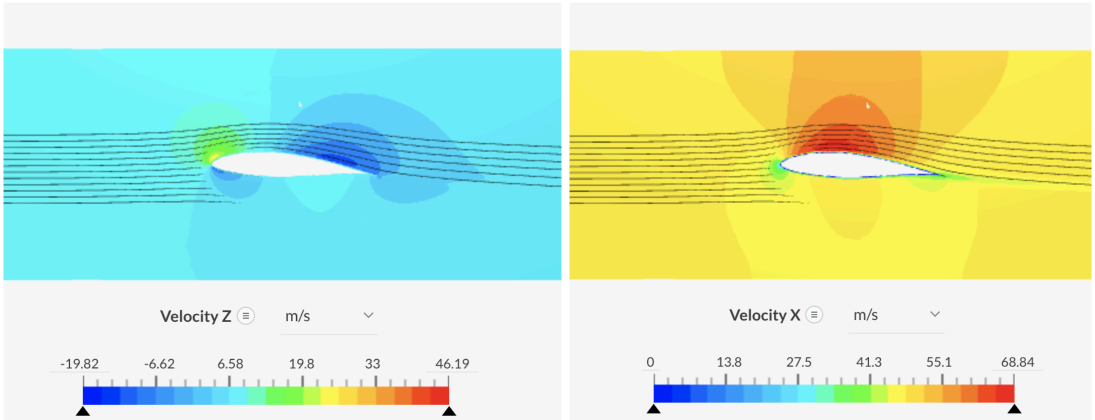

Figure 4: SimScale Side view of flow velocity contours in the X and Z directions

The pressure contour in Figure 5 below reveals negative pressure above the wing, positive pressure below the wing, and a highly concentrated area of high positive pressure at the leading edge of the airfoil.

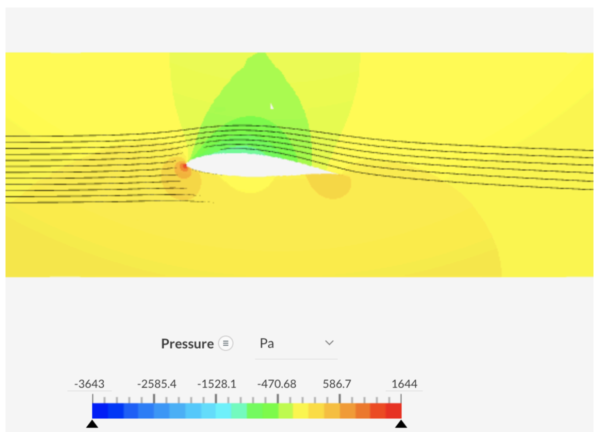

Figure 5: Simscale Side view of flow pressure contour

The features in the Z-direction velocity are caused by the Coanda effect which is the tendency for flow to follow the curvature of the surface and stay attached. This means that whenever the surface is curved upwards, the flow Z-direction velocity will be positive, and it will be negative if the surface is curved downwards. Also, at the trailing edge, the flow satisfies the Kutta Condition and exists smoothly, causing the flow to leave downwards. This requires circulation, which is an indication of lift. The X-direction velocity and the Pressure are related due to Bernouli's Principle, which says that an increase in velocity will result in a decrease in pressure. In Figure 4, there is an increase in streamwise velocity above the wing, and in the same area in Figure 5, a decrease in pressure can be seen.

TASK 5: FORCE/MOMENT COEFFICIENTS

In order to calculate the total lift and drag, we can sum up the total forces in X and Z, respectively. These forces were given by the force plots on SimScale, taken at a time where the values were steady. Equation 3 and 4 show the total drag and lift, where D is the total drag, P_X is the pressure force in the X direction, V_X is the viscous force in the X direction, L is the total lift, P_Z is the pressure force in the Z direction, and V_Z is the viscous force in the Z direction

$$D = P_X + V_X \quad [3]$$

$$D = 23.27\,N + 10.42\,N$$

$$D = 33.69\,N$$

$$L = P_Z + V_Z \quad [4]$$

$$L = 917.09\,N - 0.22\,N$$

$$L = 916.87\,N$$

The normal and axial forces can be calculated by using Equation 5 and 6 below, where α is the angle of attack.

$$N = Lcos\alpha + Dsin\alpha \quad [5]$$

$$N = (916.87\,N)cos4° + (33.69)sin4°$$

$$N = 916.99\,N$$

$$A = -Lsin\alpha + Dcos\alpha \quad [6]$$

$$A = -(916.87\,N)sin4° + (33.69)cos4°$$

$$A = -30.35\,N$$

The corresponding coefficient for each force and the moment can be calculated by dividing the force or moment by dynamic pressure and section area. Equation 7 shows the calculation for the value K, which is dynamic pressure times the section area, where Q is the dynamic pressure, S is the section area, s is the span, $\bar{c}$ is the average chord, c_R is the root chord, and c_T is the tip chord.

$$K = QS \quad [7]$$

$$Q = \frac{1}{2}\rho U_{\infty}^{2} = \frac{1}{2}(1.292\,kg/m^3)(40\,m/s)^2 = 1033.6\,Pa$$

$$S = \bar{c}s = (\frac{c_R + c_T}{2})s = (\frac{0.7294\,m + 0.30526\,m}{2})(3.048m) = 1.577\,m^2$$

$$K = (1033.6\ Pa)(1.577\ m^2) = 1629.8\ N$$

Table 1 below shows a summary of all relevant forces and moments with their corresponding coefficient. This coefficient was calculated by taking the relevant force and dividing by the K value that was found above. For the moment, an additional step was taken by also dividing by another (average) chord length. Note: the moment given by Simscale is the quarter chord moment because the origin was set there.

Table 1. Summary of all relevant forces and corresponding coefficient

Force/Moment	Magnitude	Coefficient
Lift	$L = 916.87\ N$	$C_L = \frac{L}{K} = 0.56256$
Drag	$D = 33.69\ N$	$C_D = \frac{D}{K} = 0.02607$
Normal	$N = 916.99\ N$	$C_N = \frac{N}{K} = 0.56264$
Axial	$A = -30.35\ N$	$C_A = \frac{A}{K} = -0.01862$
Moment	$M_{c/4} = -232.18\ Nm$	$C_{M_{c/4}} = \frac{M_{c/4}}{Kc} = -0.2754$

Next, the x_{CP} center of pressure can be calculated using Equation 8 below. Because it uses the moment about the quarter-chord, x_{CP} will be the distance from the quarter-chord.

$$x_{CP} = -\frac{M_{c/4}}{N} \qquad [8]$$

$$X_{CP} = -\frac{-242.18\ Nm}{916.99\ N} = 0.25320\ m$$

This can be seen in Figure 6 below. The center of pressure is used in stability analysis of aerodynamic objects. While stability is important, the center of pressure calculation is usually superfluous due to the fact that it is actually a function that changes depending on the flow condition. So as the aircraft changes speed, altitude, and angle, the center of pressure also changes. In reality, it is more productive to analyze the aircraft performance using the lift per unit span, drag per unit span, and the moment about the quarter-chord per unit span.

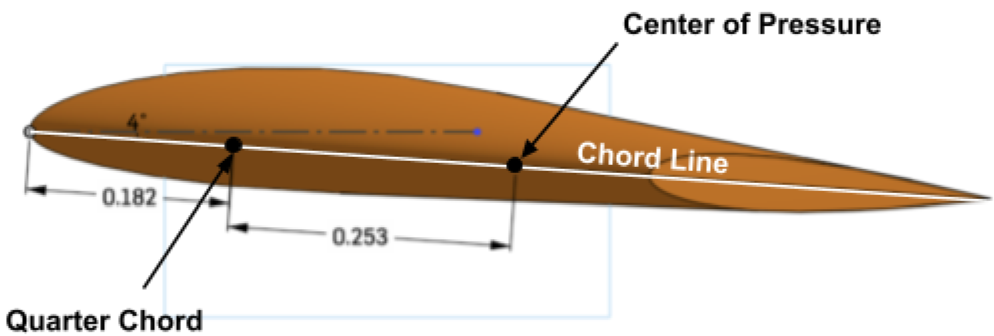

Figure 6: OnShape Side View of wing with labeled center of pressure point